

# Problem	ML Type	Technique Coverage	Why It Works Well for Training
1 Predict Patient Readmission Risk	Supervised Classification	Logistic Regression, Decision Tree, Random Forest, XGBoost	Classic healthcare prediction use case with meaningful tabular features
2 Estimate Hospital Length of Stay	Supervised Regression	Linear Regression, Random Forest Regressor, Gradient Boosting	Good for regression metrics and feature importance
3 Detect Abnormal Patient Vitals	Anomaly Detection	Isolation Forest, One-Class SVM, Z-score/IQR rules	Easy to generate normal vs abnormal vitals data
4 Cluster Patients into Risk Segments	Unsupervised Clustering	K-Means, Hierarchical Clustering, DBSCAN	Helps learners understand segmentation without labels
5 Predict No-Show for Appointments	Supervised Classification	Logistic Regression, Naive Bayes, Random Forest	Business-friendly operational ML problem
6 Identify High-Cost Claims or Billing Outliers	Anomaly Detection / Regression	Regression, Isolation Forest, Outlier Analysis	Combines payer/provider analytics with fraud/waste detection
7 Similar patient risk classification	Supervised Classification / Instance-Based Learning	KNN Classifier, distance metrics, feature scaling, hyperparameter tuning for K	Very intuitive healthcare use case: classify a new patient by comparing them with similar historical patients

TASKS

In colab or VSCode+Copilot/Claude:

Recommended Training Flow

A good progression for participants would be:

Stage Exercise

- 1 Generate synthetic dataset based on your use-case
 - 2 Perform EDA and data quality checks
 - 3 Clean missing values and outliers
 - 4 Encode categorical variables
 - 5 Split train/test data
 - 6 Train baseline model
 - 7 Train advanced model
 - 8 Evaluate using appropriate metrics
 - 9 Interpret important features
 - 10 Package model as a simple prediction function or API
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1. Predict Patient Readmission Risk

Goal: Predict whether a patient will be readmitted within 30 days after discharge.

Target variable:

readmitted_30_days: Yes/No

Synthetic dataset columns:

Column	Example
patient_age	67
gender	Male/Female
diagnosis_group	Diabetes, Cardiac, Respiratory, Renal
num_prior_admissions	3
length_of_stay_days	6
discharge_disposition	Home, Rehab, Nursing Facility

Column	Example
medication_count	8
has_followup_scheduled	Yes/No
comorbidity_score	4
readmitted_30_days	Yes/No

ML techniques covered:

Classification models, class imbalance, confusion matrix, precision, recall, F1-score, ROC-AUC, feature importance.

Training angle:

Useful for teaching why **recall** may matter more than accuracy in healthcare risk prediction.

2. Estimate Hospital Length of Stay

Goal: Predict how many days a patient is likely to stay in the hospital.

Target variable:

length_of_stay_days

Synthetic dataset columns:

Column	Example
patient_age	72
admission_type	Emergency, Elective, Urgent
diagnosis_group	Orthopedic, Cardiac, Infection
surgery_required	Yes/No
ICU_admission	Yes/No
initial_severity_score	7
num_comorbidities	4
insurance_type	Private, Government, Self-pay
length_of_stay_days	9

ML techniques covered:

Regression, feature encoding, MAE, RMSE, R^2 , residual analysis, regression model comparison.

Training angle:

Great for showing how operational teams can use ML for **bed planning and capacity forecasting**.

3. Detect Abnormal Patient Vitals

Goal: Identify patient records where vitals look abnormal or suspicious.

Target variable:

Can be optional. This can be treated as unsupervised anomaly detection.

Synthetic dataset columns:

Column	Example
heart_rate	88
systolic_bp	124
diastolic_bp	82
respiratory_rate	18
oxygen_saturation	97
temperature_celsius	37.1
age	59
ward_type	ICU, General, ER
anomaly_flag	Normal/Anomaly, optional

ML techniques covered:

Rule-based outlier detection, Z-score, IQR, Isolation Forest, One-Class SVM, visualization of anomalies.

Training angle:

Participants can inject synthetic anomalies such as:

Pattern	Example
Very low oxygen	SpO ₂ below 85
High fever	Temperature above 40°C
Extreme BP	Systolic BP above 200
Implausible records Heart rate 30 with normal ward status	
This is easy to understand and visually explain.	

4. Cluster Patients into Risk Segments

Goal: Group patients into meaningful segments such as low-risk, chronic-care, acute-risk, and high-utilization patients.

Target variable:

None.

Synthetic dataset columns:

Column	Example
age	64
BMI	31
comorbidity_score	5
annual_visits	12
prior_admissions	3
average_lab_risk_score	7
medication_count	9
chronic_condition_count	4
annual_claim_cost	185000

ML techniques covered:

K-Means, elbow method, silhouette score, PCA visualization, hierarchical clustering, cluster profiling.

Training angle:

Excellent for teaching that unsupervised ML does not “predict” a label. Learners must interpret clusters using aggregate statistics.

Example cluster interpretation:

Cluster Interpretation

Cluster 0 Young, low utilization

Cluster 1 Older, chronic conditions, moderate cost

Cluster 2 High admission, high medication, high cost

Cluster 3 High BMI and lab risk but low current utilization

5. Predict Patient Appointment No-Show

Goal: Predict whether a patient will miss a scheduled appointment.

Target variable:

no_show: Yes/No

Synthetic dataset columns:

Column	Example
patient_age	45
appointment_department	Cardiology, Dermatology, Oncology
days_between_booking_and_visit	14
previous_no_show_count	2
appointment_hour	10
weekday	Monday
reminder_sent	Yes/No
distance_from_clinic_km	12
insurance_type	Private, Government, Self-pay
no_show	Yes/No

ML techniques covered:

Binary classification, categorical encoding, train-test split, handling imbalance, threshold tuning.

Training angle:

Easy for participants to connect ML output with business action:

Prediction	Action
High no-show risk	Send additional reminder
Medium risk	Offer teleconsultation
Low risk	Standard process

This is one of the easiest use cases for end-to-end ML lifecycle training.

6. Identify High-Cost Claims or Billing Outliers

Goal: Detect unusual insurance claims or unexpectedly high treatment costs.

Target variable options:

For regression:

claim_amount

For anomaly detection:

is_outlier

Synthetic dataset columns:

Column	Example
patient_age	58
diagnosis_code_group	Cardiac, Oncology, Ortho, Neuro
procedure_type	Surgery, Imaging, Consultation
hospital_type	Primary, Secondary, Tertiary
length_of_stay_days	5
ICU_days	2
number_of_procedures	3
medication_cost	42000

Column	Example
lab_cost	11000
claim_amount	245000
is_outlier	Yes/No, optional

ML techniques covered:

Regression, residual-based anomaly detection, Isolation Forest, cost prediction, outlier ranking.

Training angle:

Participants can first predict expected claim amount, then flag claims where:

`actual_claim_amount >> predicted_claim_amount`

This introduces a very practical concept: **model-based anomaly detection**.

7. Predict Patient Risk Category Using Similar Patient History

Problem:

Given a new patient's structured profile, classify them into a risk category by comparing them with similar historical patients.

Target variable:

risk_category: Low / Medium / High

Synthetic dataset columns:

Column	Example
age	62
BMI	31
systolic_bp	148
diastolic_bp	92
fasting_glucose	132
cholesterol	245
comorbidity_score	4

Column	Example
prior_admissions	2
medication_count	6
smoking_status	Current / Former / Never
activity_level	Low / Medium / High
risk_category	Low / Medium / High

Why KNN fits well:

KNN can classify a new patient by finding the “K” most similar patients based on age, vitals, lab values, comorbidities, and utilization history.

Example explanation:

This patient is classified as **High Risk** because 7 out of the 10 most similar historical patients were also high risk.

That makes KNN easy to explain to non-technical healthcare participants.

Techniques covered

Concept	What learners practice
Distance-based learning	Similarity between patients
Feature scaling	Why StandardScaler/MinMaxScaler is critical
Choosing K	Compare K = 3, 5, 7, 11
Classification	Predict Low/Medium/High risk
Model evaluation	Accuracy, F1-score, confusion matrix
Interpretability	Show nearest patient records
Limitations	KNN becomes slower and weaker with many noisy features